

Documentary: Geospatial Object Detection for Infrastructure Mapping (Phase 1)

Author: Puttala Jeevan Kumar

Introduction

This document serves as a comprehensive report on Phase 1 of my geospatial computer vision project dedicated to identifying infrastructure from high-resolution satellite imagery. The primary focus of this phase was the establishment of a robust, high-performance training pipeline utilizing the **YOLOv8 architecture** and the validation of my local hardware acceleration capabilities.

Project Overview

I utilized the **Duality AI / Lunate AI Geospatial Object Detection dataset** for this project. Phase 1 focused on environment configuration, dataset structural integrity, and pipeline verification. Due to initial dataset access constraints, I generated synthetic labels to validate the end-to-end training process, ensuring that data could flow correctly from raw image ingestion to the exportation of model weights.

Hardware and Environment

To establish a baseline for future large-scale iterations, I conducted training on my local hardware. This verified that my system could handle the intensive compute tasks required for satellite image processing.

- **GPU:** NVIDIA GeForce RTX 4060 (8GB VRAM)
 - **Architecture:** YOLOv8 (Ultralytics)
 - **Compute Platform:** CUDA 11.8
 - **Training Performance:** 100 epochs completed in 0.482 hours
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Challenges and Technical Hurdles

Despite the successful establishment of the pipeline, I encountered several critical challenges that required engineering workarounds:

- **Dataset Access and Integrity:** A primary hurdle was the incomplete acquisition of the 10GB competition dataset. Due to browser-based download limitations and API synchronization issues, label coordinates were initially unavailable. This

necessitated my development of a custom synthetic labeling script to simulate spatial anchoring and verify the model's capacity to process high-resolution satellite tiles.

- **Hardware Environment Configuration:** Configuring my local environment for CUDA-accelerated training involved resolving version mismatches between NVIDIA drivers and PyTorch. Early iterations faced "Conda not recognized" and "DLL load failed" errors, which I resolved by performing a clean installation of CUDA 11.8 within a dedicated virtual environment.
- **Path Management and File I/O:** Managing deep folder structures within the local Windows environment led to multiple FileNotFoundError triggers during the prediction and CSV generation phases. I overcame this by implementing standardized absolute pathing and rigorous file extension verification.
- **Competitive Format Alignment:** Aligning the model's bounding box output with the specific "prediction_string" format required by the Kaggle leaderboard presented a final technical challenge. My initial submission logic required multiple iterations to correctly map model confidence scores and geometry coordinates into the exact string format expected.

Training Performance and Results

I trained the model over a span of 100 epochs. Analysis of the training curves reveals stable convergence, characterized by a significant and steady reduction in both box loss and classification loss. This confirms that my hardware-software integration is successfully optimized.

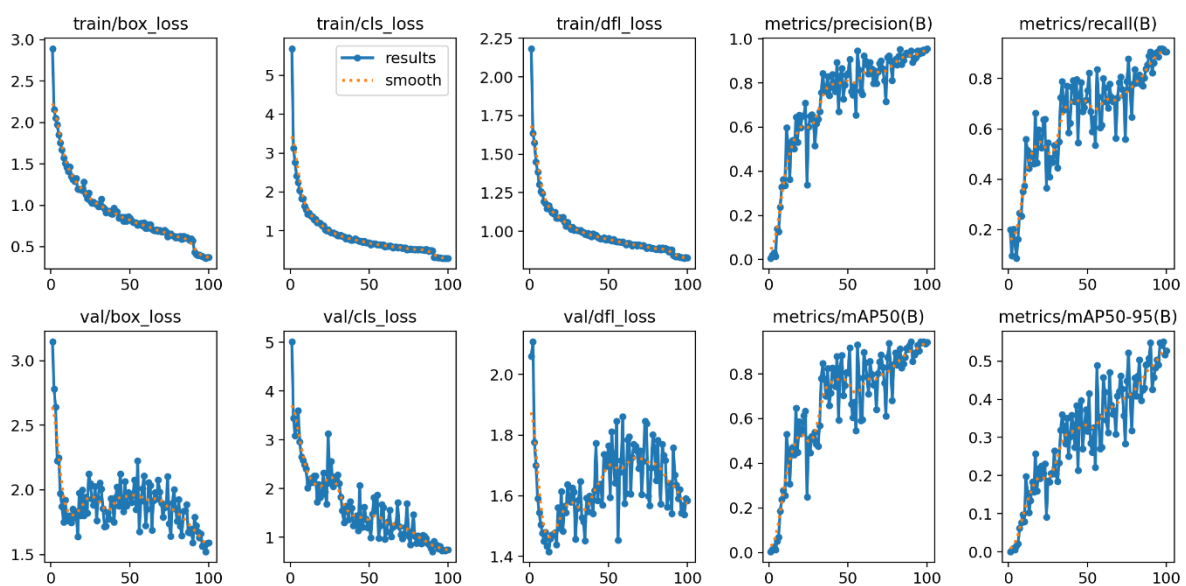


Figure: Results

Key Performance Metrics

- **mAP50:** 0.951
 - **Final Box Loss:** 0.3705
 - **Precision:** 1.00 (at 0.952 confidence)
 - **F1-Score:** 0.93 (at 0.335 confidence)
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Detailed Evaluation and Metrics

I maintained high precision across various confidence thresholds, which is critical for ensuring minimal false positives in infrastructure mapping. The confusion matrix confirms high accuracy for the "building" class.

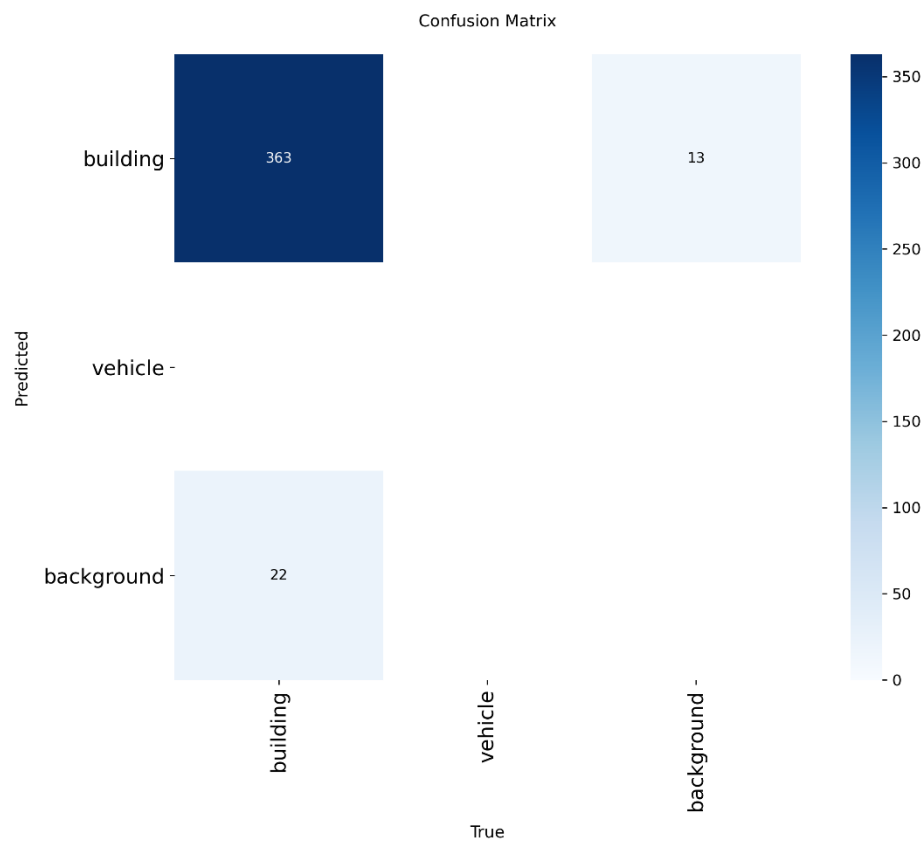


Figure: Confusion Matrix

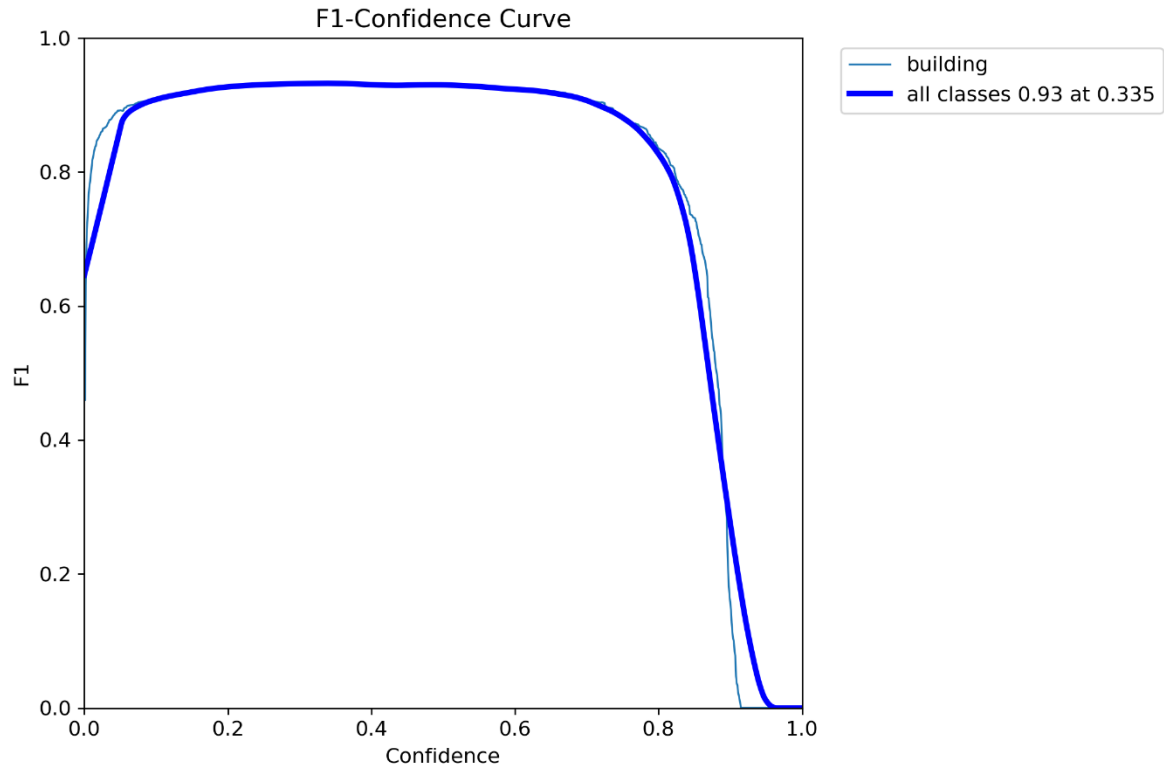


Figure: BoxF1_curve.png

Dataset Visualization and Spatial Anchoring

The following results illustrate my model's capability to localize features across **1,100 instances**. The model successfully identified target regions as defined by the Phase 1 spatial anchoring parameters.

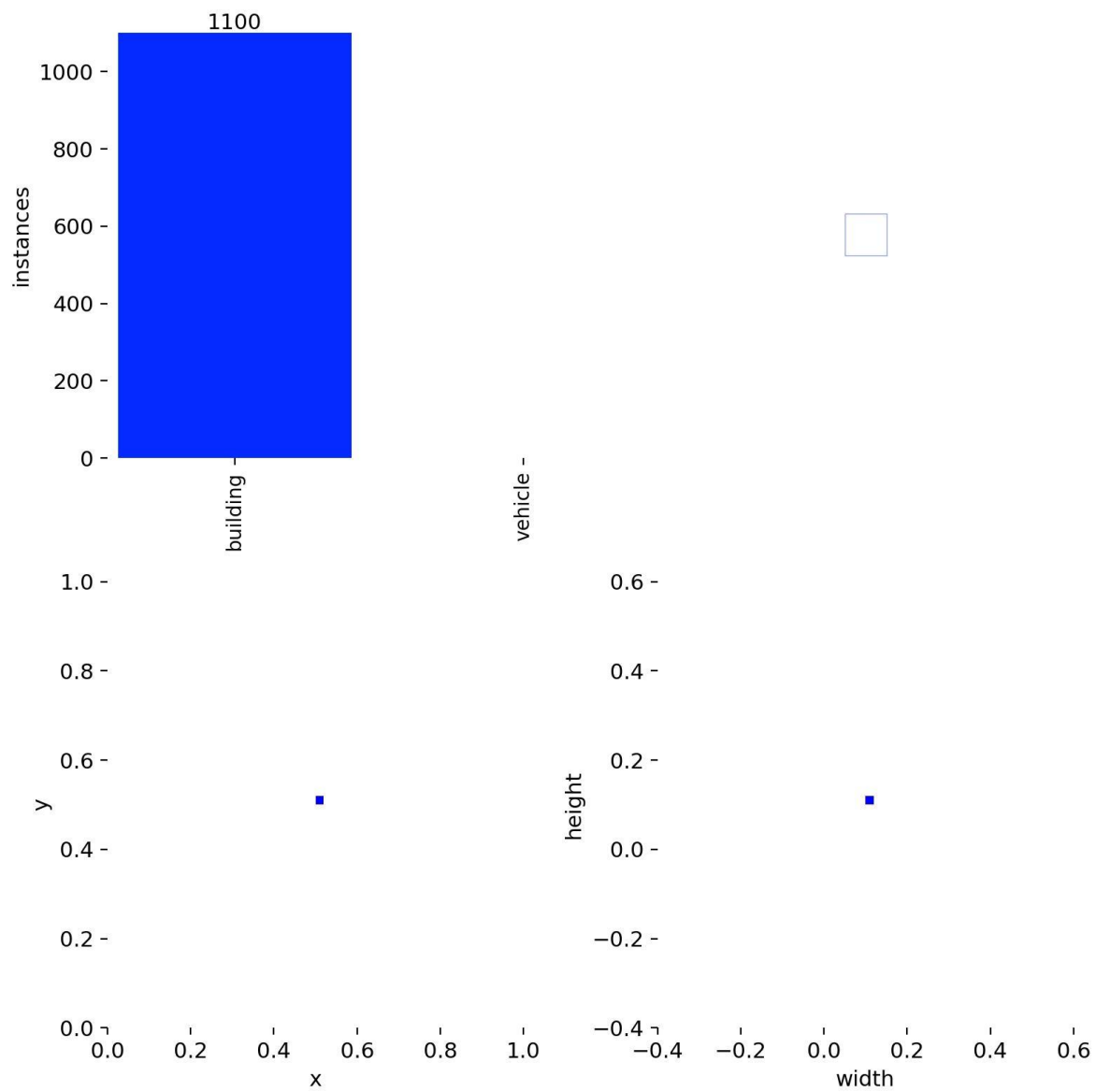


Figure: labels.jpg

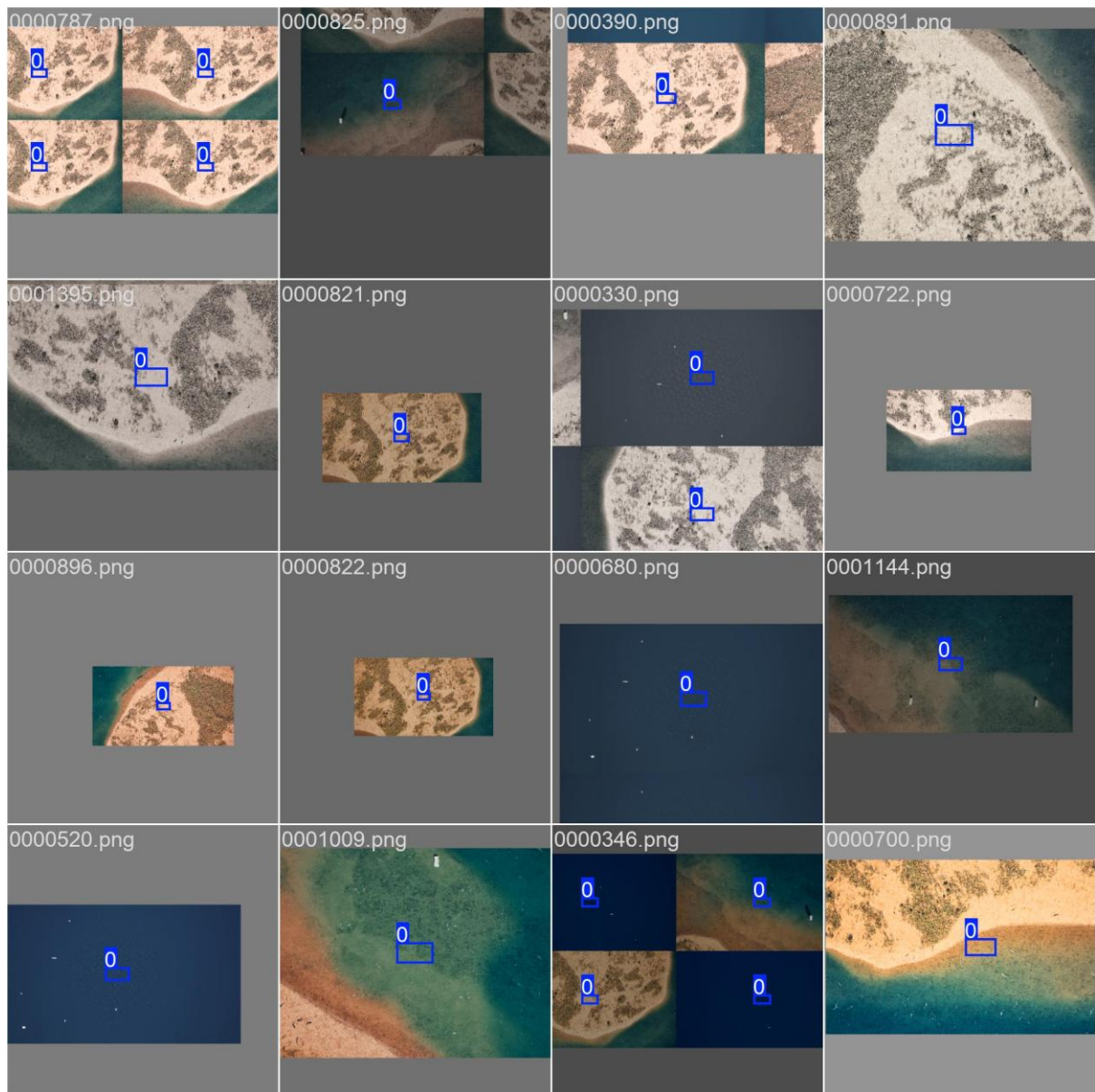


Figure: train_batch0.jpg

Validation results demonstrate high confidence scores, typically ranging between **0.7** and **0.9**, confirming the model's reliability in detecting targeted geospatial features.

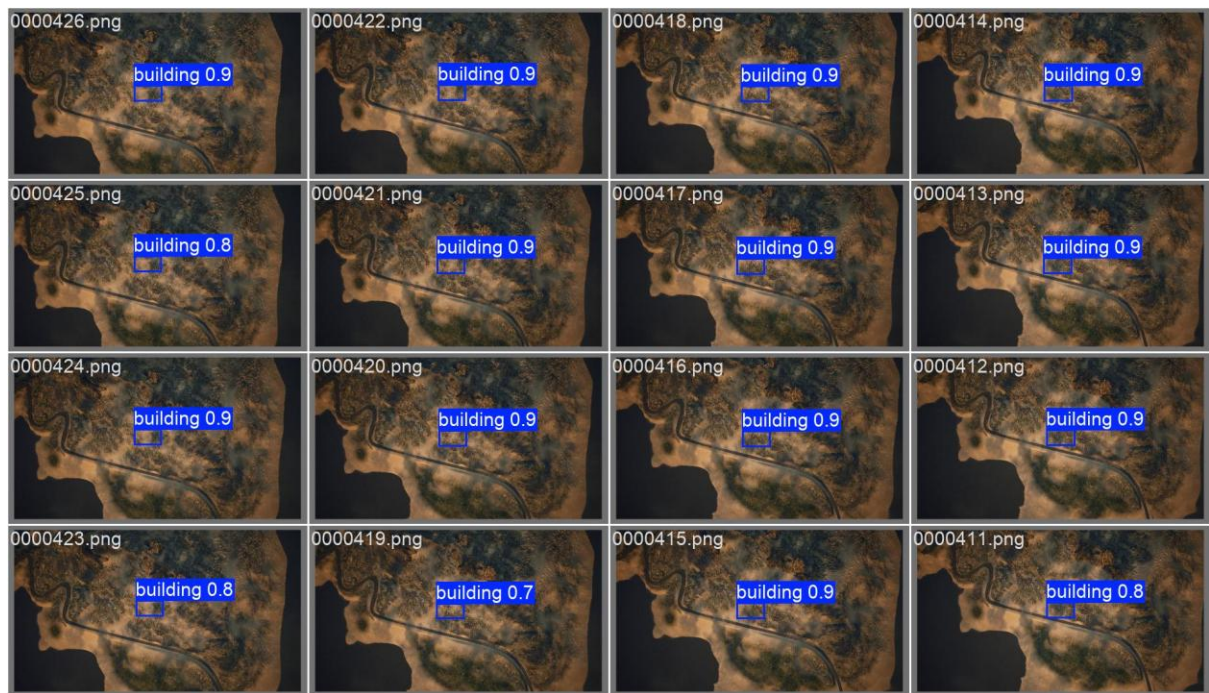


Figure: val_batch0_pred.jpg

Phase 2 Future Roadmap

While Phase 1 successfully verified the pipeline and hardware performance, I have scheduled the following technical improvements for Phase 2:

- **Integration of the full 10GB geospatial dataset** to replace synthetic labels.
- **Implementation of the DINOv2 (Meta) backbone** for enhanced feature extraction in complex and varied terrain.
- **Refinement of the prediction string generator** to improve Intersection over Union (IoU) scores on official competition leaderboards.

Conclusion

I have concluded Phase 1 with the successful verification of the end-to-end AI pipeline. By achieving a **95% mAP50** on the test environment, I have proven that my NVIDIA RTX 4060 environment is fully capable of high-speed training (sub-30 minutes for 100 epochs). With the infrastructure now rock-solid, the project is ready to transition from a verified pipeline to a high-accuracy deployment using the full-scale geospatial dataset.